

# Data preparation in Data Science

- Good data preparation allows for efficient analysis, limits errors and inaccuracies that can occur to data during processing, and makes all processed data more accessible to users.
- Majority of a data scientist's time around 80% is actually spent preparing data because the data is seldom in any order to actually perform analysis.
- Using Python and the various libraries it provides makes the task a lot simpler, faster, and more efficient.

## What is Data Preparation?

- Data preparation is the process of cleaning and transforming raw data prior to processing and analysis. It is an important step prior to processing and often involves reformatting data, making corrections to data and the combining of data sets to enrich data.
- Data preparation is often a lengthy undertaking for data professionals or business users, but it is essential as a prerequisite to put data in context in order to turn it into insights and eliminate bias resulting from poor data quality.

## Benefits of Data Preparation

- Fix errors quickly
- Produce top-quality data
- Make better business decisions

## Data Preparation Steps

**data preparation process vary by industry, organization and need, but the framework remains largely the same.**

1. **Gather data**- The data preparation process begins with finding the right data. This can come from an existing data catalog or can be added ad-hoc.
2. **Discover and assess data**- After collecting the data, it is important to discover each dataset. This step is about getting to know the data and understanding what has to be done before the data becomes useful in a particular context.
3. **Cleanse and validate data**- Cleaning up the data is traditionally the most time consuming part of the data preparation process. Important tasks here include:
  - Removing extraneous data and outliers.

- Filling in missing values.
- Conforming data to a standardized pattern.
- Masking private or sensitive data entries.

Once data has been cleansed, it must be validated by testing for errors in the data preparation process up to this point. Often times, an error in the system will become apparent during this step and will need to be resolved before moving forward.

4. **Transform and enrich data**- Transforming data is the process of updating the format or value entries in order to reach a well-defined outcome, or to make the data more easily understood by a wider audience. Enriching data refers to adding and connecting data with other related information to provide deeper insights.
5. **Store data**- The prepared data is then stored in a data warehouse or other repository and made available for use.

## Data Cleaning

- Data cleaning is one of the important parts of machine learning. It plays a significant part in building a model. It surely isn't the fanciest part of machine learning and at the same time, there aren't any hidden tricks or secrets to uncover. However, the success or failure of a project relies on proper data cleaning. Professional data scientists usually invest a very large portion of their time in this step because of the belief that "Better data beats fancier algorithms".
- If we have a well-cleaned dataset, there are chances that we can get achieve good results with simple algorithms also, which can prove very beneficial at times especially in terms of computation when the dataset size is large. Obviously, different types of data will require different types of cleaning. However, this systematic approach can always serve as a good starting point.
- **Steps Involved in Data Cleaning**
- Data cleaning is a crucial step in the machine learning (ML) pipeline, as it involves identifying and removing any missing, duplicate, or irrelevant data. The goal of data cleaning is to ensure that the data is accurate, consistent, and free of errors, as incorrect or inconsistent data can negatively impact the performance of the ML model.
- Data cleaning, also known as data cleansing or data preprocessing, is a crucial step in the data science pipeline that involves identifying and correcting or removing errors, inconsistencies, and inaccuracies in the data to improve its quality and usability. Data cleaning is essential because raw data is often noisy, incomplete, and inconsistent, which can negatively impact the accuracy and reliability of the insights derived from it.
- The following are the most common steps involved in data cleaning:

### 1. Data inspection and exploration:

This step involves understanding the data by inspecting its structure and identifying missing values, outliers, and inconsistencies.

- Check the duplicate rows.

## 2. Removal of unwanted observations

This includes deleting duplicate/ redundant or irrelevant values from your dataset. Duplicate observations most frequently arise during data collection and Irrelevant observations are those that don't actually fit the specific problem that you're trying to solve.

- Redundant observations alter the efficiency to a great extent as the data repeats and may add towards the correct side or towards the incorrect side, thereby producing unfaithful results.
- Irrelevant observations are any type of data that is of no use to us and can be removed directly.

## 3. Handling missing data:

Missing data is a common issue in real-world datasets, and it can occur due to various reasons such as human errors, system failures, or data collection issues. Various techniques can be used to handle missing data, such as imputation, deletion, or substitution. We cannot just ignore or remove the missing observation. They must be handled carefully as they can be an indication of something important. The two most common ways to deal with missing data are:

- Dropping observations with missing values.
  - The fact that the value was missing may be informative in itself.
  - Plus, in the real world, you often need to make predictions on new data even if some of the features are missing!
- **Imputing the missing values** from past observations. Again, "missingness" is almost always informative in itself, and you should tell your algorithm if a value was missing.
- Fill in it automatically with
  - a global constant : e.g., "unknown", a new class?!
  - the attribute mean
  - the attribute mean for all samples belonging to the same class: smarter
  - the most probable value: inference-based such as Bayesian formula or decision tree
- Even if you build a model to impute your values, you're not adding any real information. You're just reinforcing the patterns already provided by other features.

## 4. Handling outliers:

- Outliers are extreme values that deviate significantly from the majority of the data. They can negatively impact the analysis and model performance. Techniques such as clustering, interpolation, or transformation can be used to handle outliers.

- To check the outliers, We generally use a box plot. A box plot, also referred to as a box-and-whisker plot, is a graphical representation of a dataset's distribution. It shows a variable's median, quartiles, and potential outliers. The line inside the box denotes the median, while the box itself denotes the interquartile range (IQR). The whiskers extend to the most extreme non-outlier values within 1.5 times the IQR. Individual points beyond the whiskers are considered potential outliers. A box plot offers an easy-to-understand overview of the range of the data and makes it possible to identify outliers or skewness in the distribution.

## 5. Data transformation

Data transformation involves converting the data from one form to another to make it more suitable for analysis. Techniques such as normalization, scaling, or encoding can be used to transform the data.

- Smoothing: remove noise from data. It includes binning , regression and clustering
- Aggregation: summarization. E.g annual sale amount can be generated from monthly sale and data cube construction for analysis at multiple abstraction level
- Concept hierarchy generation for nominal data: where attributes such as street can be generalized to higher-level concepts, like city or country
- Normalization: scaled to fall within a small, specified range
  - min-max normalization
  - z-score normalization
  - normalization by decimal scaling
- Attribute/feature construction
  - New attributes constructed from the given ones

## Advantages of Data Cleaning:

1. Improved model performance: Data cleaning helps improve the performance of the ML model by removing errors, inconsistencies, and irrelevant data, which can help the model to better learn from the data.
2. Increased accuracy: Data cleaning helps ensure that the data is accurate, consistent, and free of errors, which can help improve the accuracy of the ML model.
3. Better representation of the data: Data cleaning allows the data to be transformed into a format that better represents the underlying relationships and patterns in the data, making it easier for the ML model to learn from the data.
4. Improved data quality: Data cleaning helps to improve the quality of the data, making it more reliable and accurate. This ensures that the machine learning models are trained on high-quality data, which can lead to better predictions and outcomes.
5. Improved data security: Data cleaning can help to identify and remove sensitive or confidential information that could compromise data security. By eliminating this

information, data cleaning can help to ensure that only the necessary and relevant data is used for machine learning.

## **Disadvantages of Data Cleaning:**

1. Time-consuming: Data cleaning can be a time-consuming task, especially for large and complex datasets.
2. Error-prone: Data cleaning can be error-prone, as it involves transforming and cleaning the data, which can result in the loss of important information or the introduction of new errors.
3. Limited understanding of the data: Data cleaning can lead to a limited understanding of the data, as the transformed data may not be representative of the underlying relationships and patterns in the data.
4. Data loss: Data cleaning can result in the loss of important information that may be valuable for machine learning analysis. In some cases, data cleaning may result in the removal of data that appears to be irrelevant or inconsistent, but which may contain valuable insights or patterns.
5. Cost and resource-intensive: Data cleaning can be a resource-intensive process that requires significant time, effort, and expertise. It can also require the use of specialized software tools, which can add to the cost and complexity of data cleaning.
6. Overfitting: Overfitting occurs when a machine learning model is trained too closely on a particular dataset, resulting in poor performance when applied to new or different data. Data cleaning can inadvertently contribute to overfitting by removing too much data, leading to a loss of information that could be important for model training and performance.

## **Slicing and Dicing(Practically it is covered in the exploring pandas)**

Slicing and Dicing refers to a way of segmenting, viewing and comprehending data in a database. Large blocks of data is cut into smaller segments and the process is repeated until the correct level of detail is achieved for proper analysis. Therefore slicing and dicing presents the data in new and diverse perspectives and provides a closer view of it for analysis. For example a report is showing annual performance of a particular product. If we want to view the quarterly performance, we can use slicing and dicing strategy to drill down to the quarterly level.

## **Data Wrangling and its Techniques**

- Data Wrangling is the process of gathering, collecting, and transforming raw data into another format for better understanding, decision-making, accessing, and analysis in less time. Data Wrangling is also known as Data Munging.

- Data wrangling can be defined as the process of cleaning, organizing, and transforming raw data into the desired format for analysts to use for prompt decision-making. Also known as data cleaning or data munging, data wrangling enables businesses to tackle more complex data in less time, produce more accurate results, and make better decisions. The exact methods vary from project to project depending upon your data and the goal you are trying to achieve. More and more organizations are increasingly relying on data wrangling tools to make data ready for downstream analytics.

- **Importance Of Data Wrangling**

- Data Wrangling is a very important step in a Data science project. The below example will explain its importance:
- Books selling Website want to show top-selling books of different domains, according to user preference. For example, if a new user searches for motivational books, then they want to show those motivational books which sell the most or have a high rating, etc. But on their website, there are plenty of raw data from different users. Here the concept of Data Munging or Data Wrangling is used. As we know Data wrangling is not by the System itself. This process is done by Data Scientists. So, the data Scientist will wrangle data in such a way that they will sort the motivational books that are sold more or have high ratings or user buy this book with these package of Books, etc. On the basis of that, the new user will make a choice. This will explain the importance of Data wrangling.

- **Benefits of Data Wrangling**

- Data wrangling helps to improve data usability as it converts data into a compatible format for the end system.
- It helps to quickly build data flows within an intuitive user interface and easily schedule and automate the data-flow process.
- Integrates various types of information and their sources (like databases, web services, files, etc.)
- Help users to process very large volumes of data easily and easily share data-flow techniques. **Data Wrangling in Python** Data Wrangling is a crucial topic for Data Science and Data Analysis. Pandas Framework of Python is used for Data Wrangling. Pandas is an open-source library in Python specifically developed for Data Analysis and Data Science. It is used for processes like data sorting or filtration, Data grouping, etc.

**Data wrangling in Python deals with the below functionalities:**

1. Data exploration: In this process, the data is studied, analyzed, and understood by visualizing representations of data.
2. Dealing with missing values: Most of the datasets having a vast amount of data contain missing values of NaN, they are needed to be taken care of by replacing them with mean, mode, the most frequent value of the column, or simply by dropping the row having a NaN value.
3. Reshaping data: In this process, data is manipulated according to the requirements, where new data can be added or pre-existing data can be modified.

4. Filtering data: Some times datasets are comprised of unwanted rows or columns which are required to be removed or filtered
5. Other: After dealing with the raw dataset with the above functionalities we get an efficient dataset as per our requirements and then it can be used for a required purpose like data analyzing, machine learning, data visualization, model training etc.

## Data Normalization

- Data normalization is a technique used in data science to transform the values of a dataset into a common scale. This is important because many machine learning algorithms are sensitive to the scale of the input features and can produce better results when the data is normalized.
  - Normalization is used to scale the data of an attribute so that it falls in a smaller range, such as -1.0 to 1.0 or 0.0 to 1.0. It is generally useful for classification algorithms.
  - **Need of Normalization –**
  - Normalization is generally required when we are dealing with attributes on a different scale, otherwise, it may lead to a dilution in effectiveness of an important equally important attribute (on lower scale) because of other attribute having values on larger scale. In simple words, when multiple attributes are there but attributes have values on different scales, this may lead to poor data models while performing data mining operations. So they are normalized to bring all the attributes on the same scale.
1. Min-Max normalization: This technique scales the values of a feature to a range between 0 and 1. This is done by subtracting the minimum value of the feature from each value, and then dividing by the range of the feature.
  2. Z-score normalization: This technique scales the values of a feature to have a mean of 0 and a standard deviation of 1. This is done by subtracting the mean of the feature from each value, and then dividing by the standard deviation.
  3. Decimal Scaling: This technique scales the values of a feature by dividing the values of a feature by a power of 10.
  4. Logarithmic transformation: This technique applies a logarithmic transformation to the values of a feature. This can be useful for data with a wide range of values, as it can help to reduce the impact of outliers.
  5. Root transformation: This technique applies a square root transformation to the values of a feature. This can be useful for data with a wide range of values, as it can help to reduce the impact of outliers.
  6. It's important to note that normalization should be applied only to the input features, not the target variable, and that different normalization technique may work better for different types of data and models.

**Data normalization in data science can have a number of advantages and disadvantages.**

- **Advantages:**

1. Improved performance of machine learning algorithms: Normalization can help to improve the performance of machine learning algorithms by scaling the input features to a common scale. This can help to reduce the impact of outliers and improve the accuracy of the model.
2. Better handling of outliers: Normalization can help to reduce the impact of outliers by scaling the data to a common scale, which can make the outliers less influential.
3. Improved interpretability of results: Normalization can make it easier to interpret the results of a machine learning model, as the inputs will be on a common scale.
4. Better generalization: Normalization can help to improve the generalization of a model, by reducing the impact of outliers and by making the model less sensitive to the scale of the inputs.

- **Disadvantages:**

1. Loss of information: Normalization can result in a loss of information if the original scale of the input features is important.
2. Impact on outliers: Normalization can make it harder to detect outliers as they will be scaled along with the rest of the data.
3. Impact on interpretability: Normalization can make it harder to interpret the results of a machine learning model, as the inputs will be on a common scale, which may not align with the original scale of the data.
4. Additional computational costs: Normalization can add additional computational costs to the data mining process, as it requires additional processing time to scale the data.
5. In conclusion, data normalization can have both advantages and disadvantages. It can improve the performance of machine learning algorithms and make it easier to interpret the results. However, it can also result in a loss of information and make it harder to detect outliers. It's important to weigh the pros and cons of data normalization and carefully assess the risks and benefits before implementing it.

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